
Homework 4

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MA 564 Section A1

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Sorry for not stapling my past two homeworks, I have had difficulty finding a stapler to use.

Problem 1

Note that each element of the Cantor set can be identified with a sequence $\{a_i\}_{i=1}^{\infty}$, with notation as in the handout. We can then define a corresponding sequence $\{b_i\}_{i=1}^{\infty}$, where $b_i = 0$ if $a_i = 0$, and $b_i = 1$ if $a_i = 2$. We can then define a map $\varphi : \mathcal{C} \rightarrow [0, 1]$ as the function

$$\sum_{i=1}^{\infty} a_i 3^{-i} \mapsto \sum_{i=1}^{\infty} b_i 2^{-i}.$$

Now we show surjectivity. Let $x \in [0, 1]$. Define the series $\{x_i\}_{i=1}^{\infty}$ such that

$$x_i := \begin{cases} 1, & x \geq 2^{-i} + \sum_{j=1}^{i-1} x_j 2^{-j} \\ 0, & \text{otherwise} \end{cases}.$$

It's immediately obvious that there exists a preimage of this series in $\mathcal{C}$, so it suffices to show $x_i \rightarrow x$. If $x = 0$, then we obviously have $x_i = 0$ converges to x , and if $x = 1$, then

$$\sum_{i=1}^{\infty} 2^{-i} = \frac{1/2}{1 - 1/2} = \frac{1/2}{1/2} = 1.$$

Therefore, without loss of generality let $x \in (0, 1)$. Let $U \subseteq (0, 1)$ be an open neighborhood of x , and by definition of a basis there exists r such that $B_r(x) \cap (0, 1) \subseteq U$. Without loss of generality, we can take $B_r(x) \subseteq (0, 1)$ by taking $r < \min(d(x, 0), d(x, 1))$. It suffices to show that there exists n such that $\sum_{i=1}^n x_i 2^{-i} \in B_r(x)$, and that

$$d\left(\sum_{i=1}^n x_i 2^{-i}, x\right) \leq d\left(\sum_{i=1}^{n+1} x_i 2^{-i}\right)$$

for all n .

First, note that by definition, $d(x_1, x) \leq 2^{-1}$. Now suppose that for some n ,

$$d\left(\sum_{i=1}^n x_i 2^{-i}, x\right) \leq 2^{-n}.$$

If we also have that

$$d\left(\sum_{i=1}^n x_i 2^{-i}, x\right) \leq 2^{-n-1},$$

then by definition $x_{n+1} = 0$, and we are done. Otherwise, we have $x_{n+1} = 1$, and noting that the partial sum will always be less than x by definition, we have

$$\begin{aligned}
2^{-n} \geq x - \sum_{i=1}^n x_i 2^{-i} > 2^{-n-1} &\implies 2^{-n} - 2^{-n-1} \geq x - \sum_{i=1}^{n+1} x_i 2^{-i} > 2^{-n-1} - 2^{-n-1} \\
&\implies 2^{-n} (1 - 2^{-1}) \geq d\left(\sum_{i=1}^{n+1} x_i 2^{-i}, x\right) > 0 \\
&\implies 2^{-n} (2^{-1}) \geq d\left(\sum_{i=1}^{n+1} x_i 2^{-i}, x\right) > 0 \\
&\implies 2^{-n-1} \geq d\left(\sum_{i=1}^{n+1} x_i 2^{-i}\right) > 0.
\end{aligned}$$

By induction, we thus have

$$d\left(\sum_{i=1}^n x_i 2^{-i}, x\right) \leq 2^{-n}$$

for all n . It is easily shown that for all $r \in \mathbb{R}^+$, there exists n such that $r > 2^{-n}$. Therefore, there exists some n such that

$$\sum_{i=1}^n x_i 2^{-i} \in B_r(x).$$

Second, by our previous proof note that for any n , we have either

$$d\left(\sum_{i=1}^n x_i 2^{-i}, x\right) = d\left(\sum_{i=1}^{n+1} x_i 2^{-i}\right),$$

and thus we have

$$d\left(\sum_{i=1}^n x_i 2^{-i}, x\right) \geq d\left(\sum_{i=1}^{n+1} x_i 2^{-i}\right),$$

or we instead have

$$2^{-n} \geq d\left(\sum_{i=1}^n x_i 2^{-i}, x\right) \geq 2^{-n-1} \geq d\left(\sum_{i=1}^{n+1} x_i 2^{-i}\right).$$

Therefore, the distances are decreasing as $n \rightarrow \infty$. Therefore, there exists N such that $N \leq n$ implies the partial sum

$$\sum_{i=1}^n x_i 2^{-i} \in B_r(x).$$

Since $B_r(x)$ exists (WLOG) as a subset of any open set U , we have that the sequence of partial sums converges to x , and thus

$$\sum_{i=1}^{\infty} x_i 2^{-i} = x.$$

Before we can show continuity, we must show the following important lemma.

Lemma 1. *If $x, y \in \mathcal{C}$, then $x < y$ if and only if $\varphi(x) \leq \varphi(y)$.*

Proof. Let $x < y$. It follows immediately that there exists at least one i such that $x_i < y_i$. Let n be the minimum over all such i , and note that we must have $y_n = 2$ and $x_n = 0$ by definition. We also have that

$$\sum_{i=n+1}^{\infty} x_i 3^{-i} = \sum_{i=0}^{\infty} 3^{-n-1} x_{i+n+1} 3^{-i} \leq \sum_{i=0}^{\infty} 2 \cdot 3^{-n-1} \cdot 3^{-i} = \frac{2 \cdot 3^{-n-1}}{1 - \frac{1}{3}} = 3^{-n} < 2 \cdot 3^{-n} = y_n 3^{-n}.$$

It follows that it literally doesn't matter what the remaining x_i are from $n+1$ to ∞ , as the single y_n term in the summation is greater than the rest of the summation in x . It is obvious that this logic can be reversed to obtain the other direction of the implication. Therefore, $x < y$ in the Cantor set if and only if the largest n such that $x_n \neq y_n$ has $x_n < y_n$.

Note that the action of our definition of φ on the sequence x_i is equivalent to mapping $x_i \mapsto x_i/2$. We thus have

$$\varphi(x) = \sum_{i=1}^{\infty} x_i 2^{-i-1}.$$

We once again have

$$\sum_{i=n+1}^{\infty} x_i 2^{-i-1} = \sum_{i=0}^{\infty} 2^{-n-2} x_{n+1+i} 2^{-i} \leq \sum_{i=0}^{\infty} 2^{-n-1} \cdot 2^{-i} = \frac{2^{-n-1}}{1 - \frac{1}{2}} = 2^{-n}.$$

Also note that

$$y_n 2^{-n-1} = 2^{-n}.$$

Therefore,

$$\sum_{i=n}^{\infty} x_i 2^{-i-1} \leq y_n 2^{-n-1}.$$

The change of starting indicies from $n+1$ to n is since $x_n = 0$. This shows that $\phi(x) \leq \phi(y)$ if and only if the largest n such that $x_n \neq y_n$ has $x_n < y_n$, if and only if $x < y$ in $\mathcal{C}$. This completes the lemma. The strict $<$ relation in $\mathcal{C}$ for any nonequal series also suffices to show φ is well-defined, by showing two distinct series cannot converge to the same number, which is a nontrivial result I have until now handwaved. ■

Finally, we show continuity. Let $U \subseteq [0, 1]$ be open, and consider the preimage $\varphi^{-1}(U)$. We will split this part into two cases.

1. Suppose that $0, 1 \notin U$. We can then without loss of generality take U to be an interval (a, b) , by logic that has been used in all previous homeworks. Note that $\mathcal{U} := (\sup \varphi^{-1}(a), \inf \varphi^{-1}(b)) \cap \mathcal{C}$ is open in $\mathcal{C}$ by definition of the subspace topology, and is well-defined by lemma 1, so it suffices to show $\varphi^{-1}(a, b) = \mathcal{U}$.

Let $x \in \mathcal{U}$. Then since $\sup \varphi^{-1}(a) < x$, for any $y_a \in \varphi^{-1}(a)$, we have that $y_a < x$. Similarly, for any $y_b \in \varphi^{-1}(b)$, we have $x < y_b$. By lemma 1, we have $\varphi(y_a) \leq \varphi(x) \leq \varphi(y_b)$, and thus $a \leq \varphi(x) \leq b$. Since $x \notin \varphi^{-1}(a), \varphi^{-1}(b)$, this implies $a \neq \varphi(x)$ and $b \neq \varphi(x)$, and thus $a < \varphi(x) < b$. Therefore, $x \in \varphi^{-1}(a, b)$.

Now the converse. Let $x \in \varphi^{-1}(a, b)$. Then since $\varphi(x) \neq a, b$, we must have $\sup \varphi^{-1}(a) \leq x \leq \inf \varphi^{-1}(b)$. It suffices to show $\varphi(\sup \varphi^{-1}(a)) = a$, and similarly for b . The proof is the same for both, so I show only for a . Let $s := \inf \varphi^{-1}(a)$, and let $x < s$ be an element of $\varphi^{-1}(a)$. Then by the proof of lemma 1, there must exist a minimum n such that $s_n \neq x_n$, and in

particular $s_n > x_n$. We then have that $s_n = 2$ and $x_n = 0$, and thus the maximal value of x such that $s > x$ is

$$x = \sum_{i=1}^{n-1} x_i 3^{-i} + \sum_{i=n+1}^{\infty} 2 \cdot 3^{-i}.$$

We also clearly have that

$$s = \sum_{i=1}^{n-1} x_i 3^{-i} + 2 \cdot 3^{-n}.$$

Considering the images under φ , by lemma 1 we have

$$\sum_{i=1}^{n-1} x_i 2^{-i-1} + 2^{-n} \geq \sum_{i=1}^{n-1} x_i 2^{-i-1} + \sum_{i=n+1}^{\infty} 2^{-i}.$$

$$\begin{aligned} \varphi(s) &= \sum_{i=1}^{n-1} x_i 2^{-i-1} + 2^{-n} \\ \varphi(x) &= \sum_{i=1}^{n-1} x_i 2^{-i-1} + \sum_{i=n+1}^{\infty} 2^{-i} \\ &= \sum_{i=1}^{n-1} x_i 2^{-i-1} + \frac{2^{-n-1}}{\frac{1}{2}} \\ &= \sum_{i=1}^{n-1} x_i 2^{-i-1} + 2^{-n} \\ &= \varphi(s). \end{aligned}$$

Therefore, $\varphi(s) = \varphi(x)$, and since $\varphi(x) \in \varphi^{-1}(a)$, we have $\varphi(s) = a$. Therefore, $\varphi^{-1}(a)$ and $\varphi^{-1}(b)$ contain their sup and inf, and thus our original element x must have $\sup \varphi^{-1}(a) < x < \inf \varphi^{-1}(b)$, and thus $x \in \mathcal{U}$.

2. Suppose that either 0 or 1 is in U . Without loss of generality, take $0 \in U$. Consider the clearly open subset

$$[0, \inf \varphi^{-1}(a)) \cap \mathcal{C}.$$

The proof here is the same as for the previous case, but with slightly different notation because the interval now contains its boundary at one end, and with one extra case, being that $0 \mapsto 0$. I only realized this after writing up the other proof, and I'm not going to go change all of my notation.

Therefore, φ is continuous, as the preimage of every open set is open.

Problem 2

Consider $\mathbb{Q} \subseteq \mathbb{R}$ with the subspace topology. Since $\mathbb{R}$ and by extension $\mathbb{Q}$ are Hausdorff, singletons are open by the first homework. Suppose there exists a connected component that has at least two points $a, b \in A$. If it is connected with respect to the subspace topology $\mathbb{Q} \subseteq \mathbb{R}$, it must be connected in $\mathbb{R}$, and thus by a simple extension of a theorem done in class it must be of the form $[a, b]$. However, this contains uncountably many points, not two, a contradiction. In particular, since $\mathbb{Q}$ is countable, this generalizes to any non-singleton subset. Therefore, only singletons are open. Since singletons are not open, we are done.

Problem 3

Let $X \subseteq \mathbb{N}$ be open. Then it has finite complement. Suppose that $A, B \subseteq X$ are disjoint open sets such that $A \cup B = X$. Suppose for contradiction they are both nonempty. Since they are disjoint, $A \subseteq B^c$. Since A has finite complement, it must be infinite, as $\mathbb{N}$ is infinite and

$$|A| = |X \setminus A^c| = |X| \setminus |A^c| = |X|,$$

since $|A^c|$ is finite. However, since B^c is finite, and $A \subseteq B^c$, it follows that $|A| \leq |B^c|$, and thus A is finite, a contradiction. Therefore, either A or B is empty, and every open subset $X \subseteq \mathbb{N}$ is connected. Therefore, $\mathbb{N}$ is locally connected.

Let $X \subseteq \mathbb{N}$ be open, and suppose that $f : [0, 1] \rightarrow X$ is continuous. It suffices to show f is constant, as we then have $f(0) = f(1)$, and thus for any $x \neq y$, there is no continuous function with $f(0) = x$ and $f(1) = y$. Such $x \neq y$ exist since X is not finite, and thus cannot be a singleton.

Lemma 2. *If $U \subseteq B$ is closed and $f : A \rightarrow B$ is continuous, then $f^{-1}(U)$ is closed. In particular, $f^{-1}(A^c) = f^{-1}(A)^c$.*

Proof. Since U^c is open, $f^{-1}(U^c)$ is open. For any $x \in f^{-1}(U^c) = f^{-1}(B \setminus U)$, we must have $f(x) \in U^c$, and thus $f(x) \notin U$. Therefore, $x \in f^{-1}(B) \setminus f^{-1}(U)$. Similarly, if $x \in f^{-1}(B) \setminus f^{-1}(U)$, then $f(x) \in B \setminus U$, so $x \in f^{-1}(B \setminus U)$. It follows that

$$f^{-1}(U^c) = f^{-1}(B \setminus U) = f^{-1}(B) \setminus f^{-1}(U) = A \setminus f^{-1}(U) = f^{-1}(U)^c.$$

Therefore, $f^{-1}(U)$ is closed. ■

Let $\{x\} \subseteq \mathbb{N}$ be any singleton. Since it is finite, it must be closed, and thus $f^{-1}(x)$ is closed for any such x . For any $x \neq y$, we have $f^{-1}(x) \cap f^{-1}(y) = \emptyset$, since f is well-defined. We can thus write $[0, 1]$ as a disjoint union of closed sets

$$\bigcup_{n \in \mathbb{N}} f^{-1}(n).$$

Lemma 3. *Let A be a connected space. Then there exists no finite collection of closed, disjoint, nonempty, proper subsets $\{U_i\}_{i=1}^n$ of A such that*

$$\bigcup_{i=1}^n U_i = A.$$

Proof. Note that if $U \cup V = A$ are closed sets, then $U = V^c$ and vice versa, and thus these sets are both closed and open. By a lemma done in class, we have that $U, V = A$. The statement then follows immediately from an elementary application of induction. ■

This takes care of the case where finitely many $f^{-1}(n)$ s are nonempty, since $[0, 1]$ is connected. Suppose now for contradiction that infinitely many such fibers are connected. Since any two fibers $f^{-1}(n), f^{-1}(m)$ are disjoint, there then exist disjoint neighborhoods U_n, U_m of n and m respectively, whose existence can be verified by finding the smallest distance between any two points of $f^{-1}(n)$ and $f^{-1}(m)$, and taking unions of open balls around each element of each respective fiber, with radii no more than half of that distance.

Choose distinct $n, m \in X$. Then with notation as before, let $I_n = [a, b]$ be a connected component of $[0, 1] \setminus U_n$. Since $f^{-1}(n) \subseteq U_n$, we have that $I_n \cap U_n = \emptyset$. For all a such that $f^{-1}(a) \cap U_n = \emptyset$,

define $I_a := I_n$. Since U_n is connected, it follows that there must be infinitely many $a \in X$ such that $f^{-1}(a) \cap I_n \neq \emptyset$, otherwise by lemma 3, I_n would be disconnected, as $f^{-1}(a) \cap I_n$ is closed in I_n for all a . This process can be repeated inductively by choosing new distinct m, k such that $f^{-1}(m), f^{-1}(k) \cap I_n \neq \emptyset$, and taking I_n instead of $[0, 1]$.

This gives us a chain $\dots \subseteq I_a \subseteq I_b \subseteq I_n$ with one I_n for all $n \in X$, and by explicit construction each one has $I_n \cap f^{-1}(n) = \emptyset$. Moreover, each one is defined as an interval, and thus by the finite interval lemma

$$\bigcap_{n \in X} I_n \neq \emptyset.$$

There thus exists at least some $x \in I_n$ for all n , and thus $x \in [0, 1]$. Since the fibers over f form a partition of $[0, 1]$, there exists precisely one $n \in X$ such that $x \in f^{-1}(n)$. Since $x \in I_n$, we have $x \in I_n \cap f^{-1}(n)$, contradicting the fact that $I_n \cap f^{-1}(n) = \emptyset$, as defined in the construction of I_n .

Therefore, $[0, 1]$ cannot be partitioned into countably infinite disjoint nonempty proper closed sets, nor can it be partitioned into finitely many such sets. Therefore, all but one set must be empty, and a single fiber $f^{-1}(n)$ must be the entire interval $[0, 1]$, thus proving that f is constant.

Problem 4

For notational purposes, let $L_x := [0, 1] \times \{0\}$ be the line segment along the x -axis, let L_y be defined similarly, and let L_n be the line component $\{\frac{1}{n}\} \times [0, 1]$. Clearly each line is path connected, so I will show that for any $(1/n, x)$ and $(1/m, y)$, there exists a path. Translating to/from L_x is a subcase of this for $x = 0$, and the proof can be repeated for the case $(0, x)$ and/or $(0, y)$ to obtain the statement for L_y .

By logic used in problems 5 and 6 (which I did before this problem), translations (which can be viewed as addition) are continuous, and maps $(x, y) \mapsto (f(x), g(y))$ are continuous provided f, g are continuous. We thus have a composition of continuous maps

$$\left(\frac{1}{n}, x\right) \longmapsto \left(\frac{1}{n}, 0\right) \longmapsto \left(\frac{1}{m}, 0\right) \longmapsto \left(\frac{1}{m}, y\right).$$

Therefore, a path exists, and the set is path connected.

To show the set is not locally path connected, consider the subset $A := ([0, 1] \times [1/3, 2/3]) \cap X$, and consider the point $x = (0, 1/2)$. Any open neighborhood of x can be viewed as a union of open balls intersecting with X , so it suffices to show for any $r \in \mathbb{R}^+$, the set $B_r(x) \cap X$ is not path-connected (provided it is a subset of A). Note that for all $r > 0$, since $\mathbb{Q} \subseteq \mathbb{R}$ is dense, by a lemma I proved on a previous homework there exists $q \in \mathbb{Q}$ with $0 < q < r$. We can view q as a/b , and note that $1/b < q$. It follows that $(1/b, 1/2) \in L_b$, and $L_b \cap A = \emptyset$. Since this open neighborhood is disjoint from the subset $L_y \cap A$, and since the bottom line has been removed, it is extremely obvious that there exists no continuous path between these two points, and thus A is not path-connected.

Problem 5

Since B is bounded, there exists $r \in \mathbb{R}^+$ such that $B \subseteq B_r(0)$. It follows that $B_r(0)^c \subseteq B^c$. Since $B_r(0)^c$ is unbounded, we must have that every unbounded component C_i of B^c intersects $B_r(0)^c$ at some point, since otherwise we would have $C_i \subseteq B_r(0)$, and thus C_i is bounded, a contradiction. If $B_r(0)^c$ itself is connected, then it follows that there is precisely one connected component of B^c , since containment of a point within a connected set is an equivalence relation, and since every unbounded

connected component intersects $B_r(0)^c$ somewhere, they must all be the same component by the transitive property of an equivalence relation. More precisely, if C_1, C_2 are unbounded connected components, then there exist $x_1 \in C_1 \cap B_r(0)^c$, and $x_2 \in C_2 \cap B_r(0)^c$, and thus $x_1 \sim_C x_2$, and thus C_1, C_2 fall within a single connected component. It then suffices to show $B_r(0)^c$ is connected.

We show the set in question is path-connected. Let $x, y \in B_r(0)^c$. We know from polar coordinates that we can represent x, y as points $(r_x \cos \theta_x, r_x \sin \theta_x), (r_y \cos \theta_y, r_y \sin \theta_y)$, with $r_x, r_y \geq 0$ without loss of generality. Specifically, $r_x, r_y \geq r$. I also showed in a previous homework that if f, g are continuous functions $\mathbb{R} \rightarrow \mathbb{R}$, then $x \mapsto (f(x), g(x))$ is continuous $\mathbb{R} \rightarrow \mathbb{R}^2$. However, my proof generalizes immediately to at least $\mathbb{R}^n$, and likely to any topological space. This implies that the map

$$(x, y) \mapsto (f(x), g(y))$$

is continuous, provided f, g are continuous. I also showed that multiplication by some constant $x \mapsto kx$ is continuous $\mathbb{R} \rightarrow \mathbb{R}$. Consider the map

$$(x, y) \longmapsto \begin{pmatrix} r_y \\ r_x \end{pmatrix} x, y \longmapsto \begin{pmatrix} r_y x, r_y y \\ r_x x, r_x y \end{pmatrix}.$$

Since the identity function is continuous and multiplication is continuous, each map must be continuous, and thus their composition is continuous. Therefore, the map

$$(r_x \cos \theta_x, r_x \sin \theta_x) \mapsto (r_y \cos \theta_x, r_y \sin \theta_x)$$

is continuous. Now note that by our knowledge of polar coordinates, the map $(r_y \cos \theta_x, r_y \sin \theta_x) \mapsto (r_y \cos \theta_y, r_y \sin \theta_y)$ corresponds to some arbitrary rotation. Any rotation in $\mathbb{R}^2$ by an angle φ can be given by matrix multiplication with some rotation matrix $R \in \text{SO}_2(\mathbb{R})$:

$$\begin{pmatrix} \cos \varphi & \sin \varphi \\ -\sin \varphi & \cos \varphi \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos \varphi x + \sin \varphi y \\ -\sin \varphi x + \cos \varphi y \end{pmatrix}.$$

This can be viewed as the composition

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} \cos \varphi x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} \cos \varphi x \\ \cos \varphi y \end{pmatrix} \longmapsto \begin{pmatrix} \cos \varphi x + \sin \varphi y \\ \cos \varphi y \end{pmatrix} \longmapsto \begin{pmatrix} \cos \varphi x + \sin \varphi y \\ \cos \varphi y - \sin \varphi x \end{pmatrix}$$

Since each map is a combination of the identity and either addition or multiplication, all of which are continuous (provided x, y are fixed, which they can be in our example since we only care about two specific points), we have that the entirety of the composition is continuous, and thus any arbitrary rotation is continuous. This allows us to map $\theta_x \mapsto \theta_y$. Therefore, there exists a path

$$(r_x \cos \theta_x, r_x \sin \theta_x) \longmapsto (r_y, \cos \theta_x, r_y \sin \theta_x) \longmapsto (r_y \cos \theta_y, r_y \sin \theta_y)$$

which is continuous. Moreover, since these paths always exist in $B_r(0)^c$ since $r_x, r_y \geq r$ and $B_r(0)^c$ contains its boundary as it is closed, it follows that $B_r(0)^c$ is path connected, and thus connected, and the original statement follows.

Problem 6

Since $\sin(1/x)$ is continuous for $x > 0$, we clearly have that the component $\{(x, \sin(1/x)) \mid x > 0\}$ is path-connected, and thus connected. Moreover, since line segments are connected in $\mathbb{R}$ and

there exists a trivial homeomorphism, and since homeomorphisms preserve connectedness, we have that $\{0\} \times [-1, 1]$ is connected. It suffices to show that both sets fall within the same connected component; that is, there exists x in the first set and y in the second and a connected subset containing x and y . For notational purposes, let $A_x := \{(x, \sin 1/x) \mid x > 0\}$, and let $A_y = \{0\} \times [-1, 1]$.

We can view $\sin(1/x)$ as $\sin(g(x))$ for $g(x) = 1/x$. This function is monotonically decreasing on the interval in question, so for any x_1, x_2 , we have the image of the interval $g(x_1, x_2) = (1/x_2, 1/x_1)$. Since $\sin$ is periodic with period 2π , for any x_1, x_2 such that $g(x_2) \leq 2\pi g(x_1)$, it follows that $\sin$ will attain all values $[-1, 1]$ on $(1/x_2, 1/x_1)$.

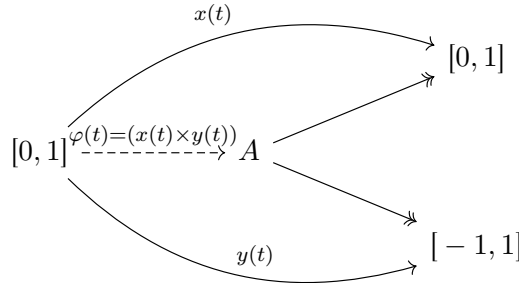
Note that for all $(0, y) \in A_y$, for all $r > 0$, there exists some x such that $d(x, y) < r$. We can then choose x' to be $x/2\pi$, and we then have that $g(x) \leq 2\pi g(x')$, and thus $\sin$ attains all values on $(1/x', 1/x)$. Since $d(x', y) < r$, it follows that there exists some $(x, \sin(1/x)) \in B_r(0, y) \cap A_x$, since $\sin$ attains all values on the interval. Therefore, for all $(0, y) \in A_y$, and for all open balls $B_r(0, y)$, the intersection $B_r(0, y) \cap A$ is nonempty.

Suppose now for contradiction that there exist two nonempty disjoint proper open sets C_1, C_2 such that $C_1 \cup C_2 = A$. Since A_x, A_y are connected, this clearly reduces to the case where $C_1 = A_x$ and $C_2 = A_y$, or more precisely, $C_1 \subseteq A_x$ and $C_2 \subseteq A_y$ are open. Since A_x is open, it can be written as a union of open balls intersecting with A , and more precisely intersecting with A_y , since $C_2 \subseteq A_y$:

$$\left(\bigcup_{i \in I} B_{r_i}(x_i) \right) \cap A_y.$$

However, as shown earlier, $B_{r_i}(x_i) \cap A_x \neq \emptyset$ for all i , and thus there exist $x \in B_{r_i}(x_i) \subseteq C_2$ such that $x \in A_x$, and thus $x \notin A_y$, a contradiction since A_x and A_y are disjoint.

Now suppose for contradiction that there exists a path $\varphi : [0, 1] \rightarrow A$ from y to x . This path exists if we constrain our choice of x, y to A_x or A_y , so choose $y_0 \in A_y$ and $x_0 \in A_x$. We can parameterize φ as a function $\varphi(t) = (x(t), y(t))$, where $x(t), y(t)$ are continuous, since they can be obtained by composing φ with the natural projection product spaces are endowed with, which is canonically continuous. More explicitly, the diagram



of continuous functions commutes.

Since we have $\varphi(0) = (0, y_0)$, and $\varphi(1) = (x_0, \sin(1/x_0))$, we must have $x(0) = 0$, $x(1) = x_0$, $y(0) = y_0$, and $y(1) = \sin(1/x_0)$. Since $\mathbb{R}$ and its open/closed subsets are Hausdorff, we have that singletons are closed, and thus by lemma 2, the preimage $x^{-1}(t_0)$ for any singleton $\{t_0\}$ is closed. Define $t_0 := \sup x^{-1}(0)$, and since the fiber of 0 is closed, it must contain its boundary (and thus its sup), so $x(t_0) = 0$.

If the function were to be continuous, we would have

$$\lim_{t \rightarrow t_0^-} \varphi(t) = \varphi(t_0) = (0, y(t_0)).$$

However, as we have shown previously, every open ball centered at an element of A_y contains elements of A_x , and if it were to converge anywhere, we would be able to define convergence at *any* element of A_y . Since limits must converge uniquely or not at all in Hausdorff spaces, we must conclude the sequence does not converge, and the limit does not exist, and thus φ is not continuous.

Problem 7

Let X be Hausdorff and $A \subseteq X$ be compact. Fix $x \in A^c$. Since X is Hausdorff, for all $y \neq x$, there exist disjoint open neighborhoods $x \in U_y$, $y \in V_y$. It follows that there exists an open cover $\{V_y\}_{y \in Y}$ of A , and since A is compact, there exists a finite subcover $\{V_{y_i}\}_{i=1}^n \subseteq \{V_y\}_{y \in A}$ of A . We can then choose a corresponding collection $\{U_{y_i}\}_{i=1}^n$ with notation as before, and it follows that the set

$$\mathcal{U}_x := \bigcap_{i=1}^n U_{y_i}$$

is open, since finite intersections of open sets are open. Since $V_{y_i} \cap U_{y_i} = \emptyset$, we have that $\mathcal{U} \cap V_{y_i} = \emptyset$ for all i . We then have

$$\mathcal{U}_x \cap \bigcup_{i=1}^n V_{y_i} = \emptyset \implies \mathcal{U}_x \subseteq \left(\bigcup_{i=1}^n V_{y_i} \right)^c \subseteq A^c.$$

With notation as before, we have

$$A^c = \bigcup_{x \in A^c} \mathcal{U}_x.$$

Since this is a union of open sets, A^c is open, and thus A is closed.